MAHAMMEDSALEEM AGASAR

Java Developer

JSpiders
Training & Development Center

ABOUT ME

I am a Full-Stack Software Developer with hands-on experience in Java-based application development. I have completed my engineering degree and strengthened my skills through structured training in core Java, SQL, and enterprise frameworks. During my training, I worked on multiple projects focused on designing, developing, and testing backend-driven applications using object-oriented principles.

I chose software development because I enjoy problem-solving and building reliable solutions from requirements. I am particularly interested in backend development with Java and Spring technologies and continuously work on improving my technical skills. My goal is to grow as a strong Java developer and contribute effectively to a professional development team.

Interesting Facts About Me

- ➢ I enjoy breaking complex problems into smaller logical steps before coding.
- ➢ I regularly practice coding challenges to improve my problem-solving ability.
- ➢ I like learning new frameworks and understanding how things work internally.I focus on writing clean, maintainable, and structured code.
- ➢ I am disciplined about continuous learning and technical growth.

Education & Certification

Bachelor's Degree - Computer Science and Engineering

Jain college of engineering and technology - Jun 2024

GPA : 7.45

Java Developer

JSPIDERS TRAINING AND DEVELOPMENT CENTER - JSP-25-1841-163904

Issued On : 02 Apr 2025

Online Transcript :
https://pdflink.to/deb62efe/

Skill Matrix

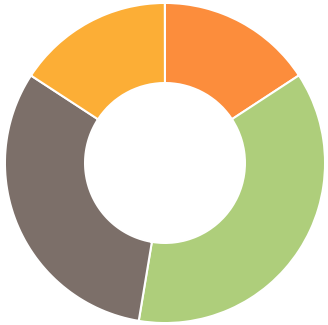
Web Technologies

- HTML (8)
- CSS (7)
- Java Script (8)
- JQuery (3)

Java

- JDBC (6)
- Java core (7)
- Collections Framework (6)
- Servlets (4)

Database



- Oracle SQL (3)
- MySql (7)
- SQL (Common for All DB) (...
- PLSQL (3)

Web Technologies

- HTML
- CSS
- Java Script
- JQuery

Java

- JDBC
- Java core
- Collections Framework
- Servlets

Database

- Oracle SQL
- MySql
- SQL (Common for All DB)
- PLSQL

TOOLS

- Git and Github (3)
- Maven (5)
- IDEs (IntelliJ/Eclipse) (7)
- Postman (6)
- STS (5)

Frameworks

- Spring (5)
- Spring Boot (6)
- Bootstrap (5)
- Thymleaf (3)
- Microservices (Basics) (3)

ORM & DB Integration

- Hibernate (5)
- JPA (5)

TOOLS

- Git and Github
- Maven
- IDEs (IntelliJ/Eclipse)
- Postman
- STS
- VS CODE
- Devops(Basic Concepts)
- JIRA

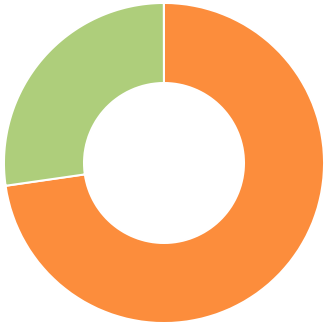
Frameworks

- Spring
- Spring Boot
- Bootstrap
- Thymleaf
- Microservices (Basics)

ORM & DB Integration

- Hibernate
- JPA

Programming Languages



Java (8)
Python (3)

Programming Languages

- Java
- Python

Projects

P2- RevPasswordManager

Application Overview

The Password Manager is a full-stack monolithic web application that allows users to securely store and manage passwords for their various online accounts. Users can create an account with a master password, generate strong random passwords, store account credentials in an encrypted vault, and manage their password entries through an intuitive web interface. The application emphasizes security through encryption, master password protection, two-factor authentication, and secure password generation.

Core Functional Requirements

As a User, I should be able to:

Authentication & Account Management

- Create an account with email, username, and master password
- Set up security questions during registration (minimum 3 questions)
- Login to my account using username/email and master password
- View my dashboard with password vault summary (total passwords, weak passwords, recently added)
- Update my profile information (name, email, phone number)
- Change master password (requires current master password verification)
- Recover forgotten master password through security questions
- Enable/disable two-factor authentication (2FA) using authenticator app simulation
- Logout from my account

Password Vault Management

- View all stored password entries in a list/grid view
- Search password entries by account name, website, or username
- Filter password entries by category (Social Media, Banking, Email, Shopping, Work, Other)
- Sort password entries by name, date added, or date modified
- View individual password entry details (requires master password re-entry)
- Add new password entry with details (account name, website URL, username/email, password, category, notes)
- Update existing password entries
- Delete password entries (with confirmation)
- Mark passwords as favorite for quick access
- View favorite passwords list

Password Generator

- Generate strong random passwords with customizable parameters:
- Password length (8-64 characters)
- Include uppercase letters
- Include lowercase letters
- Include numbers
- Include special characters
- Exclude similar characters (0, O, l, 1, etc.)
- Copy generated password to clipboard
- View password strength indicator (Weak, Medium, Strong, Very Strong)
- Save generated password directly to vault
- Generate multiple passwords and choose the preferred one

Security Features

- Re-enter master password to view sensitive information (actual passwords)
- Get a verification code for sensitive operations (email-based simulation)
- Verification code expires after single use or time limit
- View password strength analysis for stored passwords
- Receive security alerts for weak or reused passwords
- View security audit report (weak passwords, reused passwords, old passwords)
- Export password vault (encrypted backup)
- Import passwords from encrypted backup file

Security Questions Management

- Add security questions during registration
- View configured security questions (questions only, not answers)
- Update security question answers (requires master password)
- Minimum 3 security questions required

Standard Functional Scope

- Authentication & Security
- Password Encryption & Storage
- Verification & Authorization
- User Interface & Experience
- Data Management

Definition of Done

- Working Web Application Demonstration

- Code Repository Submission with
- ERD (Entity Relationship Diagram)
- Application Architecture Diagram
- README.md
- Testing Artifacts
- Security Documentation

Roles / Responsibilities

- Designed and implemented the **Vault Module** to manage secure password storage and retrieval.
- Developed PasswordEntry entity and PasswordEntryRepository using **Spring Data JPA** for database operations.
- Implemented full **CRUD operations (Add, Edit, Delete, View)** for password entries through VaultService and VaultServiceImpl.
- Integrated **AES encryption service** to encrypt passwords before saving to the database and decrypt them before displaying to users.
- Created PasswordEntryDTO and PasswordEntryMapper to separate entity and presentation layers following clean architecture principles.
- Developed VaultController (MVC) to handle vault-related HTTP requests and manage user interactions.
- Implemented vault listing features including structured display of stored credentials.
- Ensured sensitive data protection by enforcing master password verification before viewing actual passwords.

Environment / Technologies

Java, SQL, Spring Boot, JUnit4, Log4J, Git, Maven

P1- RevPassword Manager

Application Overview

The Password Manager is a secure console-based application that allows users to safely store and manage passwords for their various online accounts. Users can create an account with a master password, log in, generate strong random passwords, store account credentials, and manage their password vault. The application emphasizes security through password encryption and master password protection.

Core Functional Requirements

As a User, I should be able to:

1. Create an account to securely manage my passwords
2. Log in to my account to access the application
3. Update my profile information like name and email
4. Generate strong random passwords by providing parameters like string length, characters, numbers, etc.
5. List all the passwords with the respective account names
6. View individual password details (with master password re-entry)
7. Add new password entries for various accounts
8. Update existing passwords
9. Delete password entries
10. Search passwords by account name
11. Add security questions for account recovery
12. Get a verification code for sensitive operations which expires after use
13. Manage security questions linked to my account

Standard Functional Scope

Registered users should be able to log in, change the password and request for a forgotten password (recovered through security questions).

Definition of Done

- Working console application demonstration
- Sharing the associates' code repo for technical evaluation with:
    - ERD Diagram
    - Application Architecture Diagram (showing modular/layered design)

Roles / Responsibilities

- Designed and developed a secure, console-based password management application using Java.

- Implemented user registration and login functionality with master password validation.
- Developed features to add, update, delete, and search stored account credentials.
- Implemented password generation logic to create strong, random passwords based on user-defined parameters.
- Integrated Oracle database connectivity using JDBC for persistent data storage.
- Applied object-oriented programming principles to ensure modular and maintainable code.
- Implemented security features such as password encryption and verification for sensitive operations.
- Wrote unit tests using JUnit to validate core application functionalities.
- Used Log4J for logging application events and error handling during execution.

Environment / Technologies

Java, MySQL, JDBC, JUnit, Log4J